MMVU: Measuring Expert-Level Multi-Discipline Video Understanding

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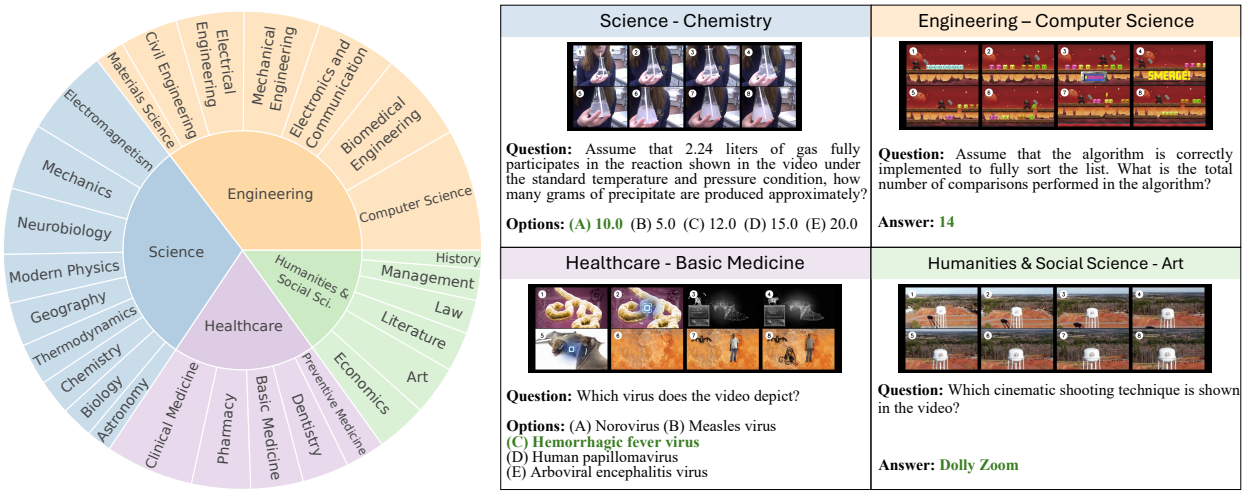

Figure 1. Overview of the MMVU benchmark, covering 27 subjects across four core disciplines. This benchmark is specifically designed to assess multimodal foundation models in expert-level, knowledge-intensive video understanding and reasoning tasks.

Abstract

We introduce MMVU, a comprehensive expert-level, multi-discipline benchmark for evaluating foundation models in video understanding. MMVU features three key advancements. First, it challenges models to apply domain-specific knowledge and perform expert-level reasoning to analyze specialized-domain videos, moving beyond the basic visual perception typically assessed in current video benchmarks. Second, each example is annotated by human experts from scratch. We implement strict data quality controls, validating that correct answers cannot be inferred solely from textual cues or answer shortcuts. Finally, each example is enriched with expert-annotated reasoning rationals and relevant domain knowledge, facilitating in-depth analysis. Our evaluation of 20 frontier models highlights a significant performance gap between the leading models and human experts. Through in-depth error analysis, case studies, and an exploration of retrieval-augmented generation methods, we offer actionable insights to guide future advancements.

1. Introduction

Foundation models have demonstrated remarkable capabilities in reasoning across various domains, yet their ability to handle expert-level knowledge remains a critical area of evaluation [60, 165]. In recent years, researchers have developed numerous benchmarks to assess these models' proficiency in specialized domains, primarily focusing on text-based reasoning [45, 60, 135, 150] and image-based contexts [89, 100, 165–167]. However, as capabilities of foundation models expand across multiple modalities, there is a significant gap in evaluating expert-level reasoning over specialized-domain *videos*. This gap is particularly concerning as video is one of the most information-rich and naturalistic modalities, and is widely used to convey complex, dynamic information in specialized fields like healthcare, engineering, and scientific research. Unlike static text or images, expert-level videos often capture temporal dynamics, procedural knowledge, and complex interactions that are essential in many specialized domains. For example, in science, expert-level and knowledge-intensive reasoning

Dataset	QA Type	Data Source	College Level?	Detailed Solution	
				Rational	Knowledge
Text					
MMLU [60]	MC	Exam, course, textbook	✓	✗	✗
MMLU-Pro [150]	MC	Datasets → Human & LLM augment	✓	✗	✗
C-Eval [63]	MC	Exam	✓	✗	✗
SciEval [135]	MC, Open	Internet,other datasets→LLM rewrite	✓	✗	✗
SciKnowEval [45]	MC, T/F, Open	Textbooks, database, other datasets → LLM rewrite	✓	✗	✓
TheoremQA [22]	MC, T/F, Open	Internet, exam → Human rewrite	✓	✗	✓
Text + Image					
VisScience [70]	MC, Open	Internet, exam, textbook	✗	✗	✗
EXAMS-V [35]	MC	Exam	✗	✗	✗
ScienceQA [99]	MC	Internet, course	✗	✓	✗
SceMQA [91]	MC, Open	Internet, exam	✗	✓	✗
CharXiv [151]	Open	arXiv paper → Human annotate	✓	✗	✗
MMSci [89]	MC	Scientific paper → LLM generate	✓	✗	✗
OlympicArena [65]	MC, T/F, Open	Olympic competitions	✓	✓	✗
MMMU [165]	MC, Open	Internet, exam, textbook	✓	17.6%	✗
CMMMU [167]	MC, T/F, Open	Internet, exam, textbook	✓	2.1%	✗
MMMU-Pro [166]	MC	MMMU → Human & LLM augment	✓	15.4%	✗
Text + Video					
MMWorld [58]	MC	Human (23.5%) / LLM (76.5%)	26.8%	✗	✗
MMVU (ours)	MC, Open	Human experts annotate from scratch	✓	✓	✓

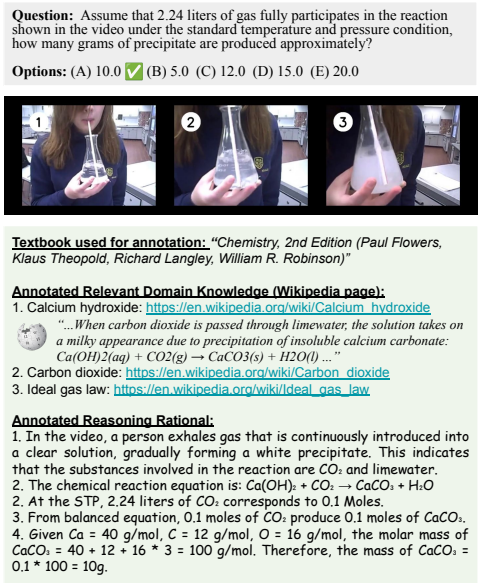

Figure 2. (Left) Comparison between MMVU and existing multi-discipline benchmarks for evaluating foundation models. In the "QA Type" column, "MC" denotes Multiple-Choice Questions, "Open" denotes Open-ended Questions, and "T/F" denotes True-False Questions. (Right) A dataset example with the subject of chemistry. MMVU has following key features: (1) **Breadth of Domain Knowledge**: We employ a textbook-guided QA annotation pipeline to ensure the wide coverage of domain knowledge within each subject (§3.2). (2) **Depth of Expert-level Reasoning**: Each example in MMVU requires models to comprehend specialized-domain video context, applying expert knowledge and reasoning (§3.2). (3) **True Visual Understanding**: Each example in MMVU is carefully validated to confirm that video comprehension is essential for accurate answering (§3.3). (4) **Support of Fine-grained Evaluation**: We provide expert-annotated solutions and the requisite knowledge for each example (§3.2), enabling more comprehensive analysis for future research (§5, §6).

might involve analyzing a chemical reaction video (Fig. 2). A model must identify key reaction stages based on subtle visual cues like color changes or the formation of precipitates, which requires integrating chemical knowledge in addition to recognizing visual patterns.

To bridge this gap, we introduce MMVU[1], a comprehensive benchmark measuring **M**ultimodal foundation models in expert-level, **M**ulti-discipline **V**ideo **U**nderstanding and reasoning. MMVU consists of 4,000 expert-annotated QA examples over 1,627 specialized-domain videos, spanning 27 subjects across four key disciplines: Science, Healthcare, Humanities & Social Sciences, and Engineering. To ensure both the breadth of domain knowledge and the depth of reasoning required for MMVU, we implement a textbook-guided data annotation process (§3.2). Expert annotators first locate key concepts from textbooks in their fields, then source relevant videos and create corresponding questions that require domain knowledge and expert-level reasoning to comprehend the videos. Each example also includes expert-annotated reasoning rationals and relevant domain knowledge, facilitating fine-grained evaluation of model performance. Thorough data quality controls are implemented to ensure high quality of MMVU (§3.3).

We conduct an extensive evaluation on MMVU, covering 20 frontier multimodal foundation models from 11 organizations. The experimental results highlight the substantial progress, especially among the latest open-source models. However, a notable performance gap persists between the highest-performing model, GPT-4o (*i.e.*, 51.9%), and human experts in an open-book setting (*i.e.*, 86.8%). We further conduct comprehensive error analysis, case studies, and an exploration of retrieval-augmented generation techniques, offering valuable insights to guide future advancements. Our contributions are summarized as follows:

- We propose MMVU, a high-quality, multi-discipline benchmark designed to assess the expert-level reasoning abilities of multimodal foundation models in comprehending specialized-domain videos. Each example is annotated from scratch by human experts, including detailed annotations such as relevant domain knowledge and reasoning rationales (§3).
- We conduct a comprehensive evaluation of 20 multimodal foundation models. The results demonstrate substantial progress, particularly among the latest open-source models. However, a significant performance gap persists between the top-performing models and human experts, in-

[1]The MMVU data and codebase are available at https://anonymous.4open.science/r/MMVU, and will be released publicly upon publication. In line with the CVPR submission policies, the Appendix is included in the supplementary materials.

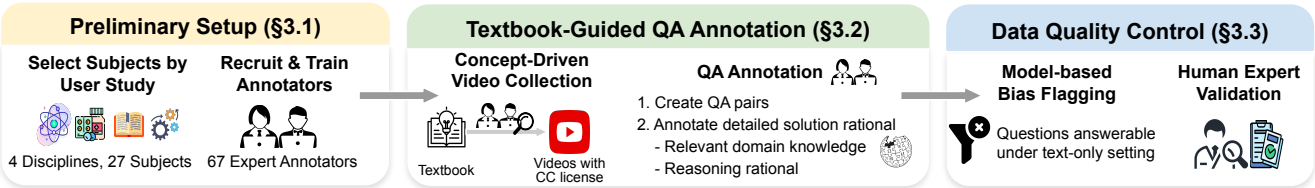

Figure 3. An overview of MMVU construction pipeline.

dicating significant room for improvement (§4).

- We perform a thorough error analysis of frontier models, featuring numerous case study examples reviewed by human experts to enhance researchers' understanding of the current models' limitations (§5.1). Additionally, we introduce an automated protocol to streamline large-scale error analysis, supporting future research efforts (§5.2).
- Our investigation into RAG-based knowledge augmentation highlights the effectiveness of incorporating domain knowledge to improve model performance (§6).

2. Related Work

Video Understanding Benchmark. Existing video understanding benchmarks primarily focus on *general-purpose* video comprehension tasks, such as action recognition [37, 59, 96, 131], captioning and description [79, 85, 137, 157, 162], grounding [24, 72, 82, 148], temporal reasoning [18, 30, 68, 73, 87, 97, 130], and long video understanding [8, 42, 112, 147, 169]. The rise of video-based foundation models [44, 64, 138, 168] has driven the development of new benchmarks that include diverse video comprehension tasks for more comprehensive evaluation [51, 74, 86, 88, 115, 161]. However, these benchmarks remain predominantly focused on natural scenes and general-purpose tasks. A significant gap persists in benchmarks targeting *expert-level* and *knowledge-intensive reasoning* over specialized-domain videos, where both visual perception and domain-specific expertise are required—especially in critical fields like healthcare, engineering, and science.

Multi-discipline Evaluation Benchmark. The rapid development of foundation models has significantly enhanced expert-level reasoning across various disciplines [53, 69, 117, 139, 164]. Early benchmarks focused on domain-specific tasks for textual domains, establishing a foundation for assessing the models' strengths and limitations in expert reasoning [22, 28, 60, 136, 150, 153, 172, 174]. More recently, benchmarks have evolved to include multimodal tasks [89, 90, 100, 151, 165–167], emphasizing visual perception and advanced reasoning with domain knowledge. However, these efforts remain largely limited to *static* images. Recent work, MMWorld [58], has made pioneering strides by incorporating videos across multiple disciplines. However, only a limited portion of its dataset (26.8%) explicitly targets domain-specific expertise, and 76.5% of the examples are generated by the GPT-4V model. Moreover, most existing benchmarks provide only the ground-truth answer, restricting researchers' ability to conduct a fine-grained evaluation of model reasoning. To address this limitation, MMVU includes expert-annotated reasoning rationales and relevant domain knowledge for each example, enabling a more nuanced assessment of expert-level reasoning. Figure 2 further distinguishes the difference between MMVU and existing multi-discipline benchmarks.

3. MMVU Benchmark

Figure 3 provides an overview of the three stages involved in constructing the MMVU benchmark, which is detailed in the following subsections.

3.1. Preliminary Setup

We first discuss the preliminary setup for data construction.

Subject Selection. To ensure a broad and accurate representation of expert-level video understanding across diverse disciplines, we conduct a user study involving 172 college and graduate students for subject selection. We ask them to curate two QA examples requiring expert-level video understanding in subjects relevant to their field of study, and provide feedback on their experiences during the curation process. Such a user study-guided approach helps us identify subjects within each discipline that may not be obvious from a top-down selection process. It also offers insights into the challenges of designing expert-level video examples, helping us design and refine the textbook-guided QA annotation process (detailed in §3.2). The authors manually analyze the collected examples and select **27 subjects** (as listed in Fig. 1) across four disciplines that align best with our benchmark's construction desiderata discussed earlier.

Expert Annotator Recruitment and Training. For each subject, we assign at least two annotators with relevant expertise. We include a total of 67 expert annotators (detailed biographies are presented in Appendix A.1), comprising 22 third- or fourth-year undergraduate students, 36 graduate students, and nine of the authors. All the annotators also

participated in our initial user study. Each annotator is required to finish a training session to learn the annotation protocol (detailed in Appendix B) before official annotation. To ensure ethical standards and participant protection, this study was reviewed and approved by Institutional Review Board (IRB) of the first author's affiliated institution.

3.2. Textbook-Guided QA Example Annotation

Developing a high-quality, multi-discipline video benchmark presents even greater challenges than those for text or image-based tasks due to the scarcity of suitable resources (*e.g.*, textbooks or exam questions). Therefore, it is crucial to establish a structured approach that ensures the quality and comprehensiveness of the benchmark. We employ a textbook-guided example annotation pipeline designed to capture both the *breadth of knowledge* and *depth of reasoning*. In brief, annotators first identify key concepts from the textbook and locate relevant videos that align with these concepts. The textbooks for each subject (listed in Appendix A.2 and cited in the references) are selected by expert annotators and are recognized as authoritative references in their respective fields. Annotators then craft QA examples and provide detailed solution rationales. We detail the annotation procedure as follows:

Concept-Driven Video Collection. Annotators are instructed to first review each chapter of the textbook to identify key concepts that inherently require dynamic visual representation, such as experimental procedures in science or mechanical operations in engineering. They then search for related videos on YouTube having Creative Commons license[2] that effectively illustrate the selected concept. To ensure the collected videos effectively challenge the model's visual reasoning capabilities, the video should be vision-intensive, requiring models to focus solely on visual information for comprehension. To this end, we ensure that audio tracks are excluded to eliminate potential shortcuts models might exploit through auditory cues; and the video should contain minimal on-screen text, as an overabundance of text may detract from the core visual understanding task. Consequently, videos such as lecture recordings, which typically include slides or text-based explanations that simplify the task of answering associated questions, are excluded.

QA Annotation. After identifying suitable videos, annotators are required to create two or three questions, either multiple-choice or open-ended. Each question is designed to test the model's expert-level reasoning by applying domain-specific knowledge to interpret the video content and derive a solution. Annotators are also required to specify the start and end timestamps of the video clip relevant to answering each question. For annotating the multi-choice question, the annotators are required to carefully craft the four distractor options to reflect common misconceptions or plausible alternatives, ensuring that models cannot easily eliminate incorrect options without reasoning over video content. Once the five options are finalized, the annotation interface randomly shuffles them.

Solution Rationale Annotation. For each annotated question, annotators must also provide detailed solution for the correct answers. These solution annotations are critical for enhancing transparency in the evaluation process and facilitating future research focused on understanding model failure modes. As shown in Fig. 2, the solution comprises two key components: (1) *relevant domain knowledge*, which includes a list of domain-specific concepts or keywords necessary for answering the question, with each concept linked to its corresponding Wikipedia page (see Appendix B.2 for details on the Wikipedia page processing). (2) *reasoning rational*, which details the step-by-step reasoning process to reach the correct answer. It must include a clear explanation of video interpretation and demonstrate how relevant knowledge is applied to answer the question.

3.3. Data Quality Control

We next discuss our methods to ensure high data quality.

Time-Based Annotation Compensation. As discussed earlier, annotating examples for MMVU can be particularly time-intensive, especially when there is limited availability of videos with Creative Commons licenses in the required subjects. To accommodate this and ensure a high-quality benchmark, we compensate annotators based on the time they spend rather than the number of examples completed, preventing them from rushing through tasks (See Appendix B.5 for annotation compensation details). On average, annotating one example takes 20 minutes and 17 seconds, while validation requires 3 minutes and 35 seconds.

Model-Based Detection of Annotation Bias. To ensure that each question in MMVU truly evaluates video-based reasoning, rather than relying on text-based dependencies or answer shortcuts [21, 166, 171], we implement a model-based detection approach using leading *open-source* models. Specifically, we test Mistral-Large, Llama-3.1-70B, and Qwen2.5-72B models in the *text-only setting*, where the model is provided with only the textual part of the question. Appendix B.3 presents the used prompt. Each model is tested five times per question, with a question considered

[2]The Creative Commons license enables reusers to distribute, remix, adapt, and build upon the material in any medium or format, so long as attribution is given to the creator. Existing video benchmarks typically utilize YouTube videos, yet often do not confine their selections to content with CC licenses, introducing potential copyright and legal concerns. We use YouTube Data API v3 to verify the license type of collected videos.

Statistics	**Value**
Total Questions	4,000
Validation Set	1,000
Test Set	3,000
Unique Videos	1,627
Video Length (Seconds, `avg/max`)	37.0 / 276
Number of Disciplines	4
Number of Subjects	27
Multiple Choice Questions	2,417
Question Length (`avg/max`)	13.7 / 66
Single Choice Length (`avg/max`)	3.8 / 18
Number of Choices per Question	**5**
Open-ended Questions	1,583
Question Length (`avg/max`)	15.2 / 34
Ground-truth Answer Length (`avg/max`)	1.8 / 7
Number of Required **Knowledge** per Question (`avg/max`)	2.9 / 6
Solution Rationale Length (`avg/max`)	47.6 / 129
Total Number of Unique Knowledge (*i.e.*, Wikipedia pages)	4,031

Table 1. Key statistics of MMVU.

"answerable" if a model correctly responds at least three times. Questions answered correctly by at least two of the models are flagged. This automated process ensures consistency in detecting potential text-based biases or unintended shortcuts that may not be immediately obvious to human.

Human Expert Validation. To ensure that the final dataset remains high-quality and meets expert-level standards without introducing unnecessary bias, each example in MMVU undergoes expert review by one of the authors or top-performing annotators to verify the accuracy of its annotations. If an example is flagged during bias detection or identified as containing annotation errors, the evaluator first attempts to revise the example. If revision is not feasible, detailed feedback is provided to the original annotator, who then revises and submits it for a second iteration. A total of 671 examples were revised during the data validation process. Among them, 96 examples were still found to be misaligned with our design criteria and were excluded from the final benchmark. Overall, $1 - \frac{671}{4{,}000+96} = 83.6\%$ of the initial examples met our design criteria without requiring revisions, indicating the high quality of initial annotation.

3.4. MMVU Benchmark Analysis

Data Statistics. Table 1 presents the key statistics of MMVU. It consists of 4,000 examples, which are randomly divided into two subsets: validation and test. The validation set contains 1,000 examples, and is intended for model development and validation. The test set contains the remaining 3,000 examples, and is reserved for standard evaluation. To avoid data contamination [38, 67], features related to the ground truth (*i.e.*, answers, relevant domain knowledge, and reasoning rationals) are withheld for the test set. Instead, we will provide an online evaluation platform on Hugging Face where researchers can benchmark their models and engage with a public leaderboard.

Human Performance. To provide a rough but informative estimate of human-level performance on MMVU, we randomly sampled 30 questions per discipline from the test set, resulting in a total of 120 questions for evaluation. Five participants—three graduate students specializing in biology, anesthesiology, and East-Asian literature, along with two of the authors—individually answered these questions. The evaluation proceeded in three phases: (1) **Closed-book Setting**: In the first phase, participants had 2.5 hours to answer questions without access to external resources. The average accuracy across the four participants was 49.7%. (2) **Open-book Setting**: In the second phase, participants were permitted to use external resources (*e.g.*, internet and textbooks) to review answers they felt uncertain about. They were not informed of the correctness of their initial responses, and a 4-hour time limit was set. This open-book approach led to an increase in average accuracy to 86.8%. (3) **Oracle Setting**: Finally, participants were required to revise each incorrect answer based on ground-truth domain knowledge and self-sourced online resources. The average accuracy after this final revision was 95.3%.

4. Main Experiment

This section discusses the experiment setup and key findings from the main experiments.

4.1. Evaluated Multimodal Foundation Models

To establish a comprehensive understanding of the challenges posed by MMVU and provide reference points for future research, we evaluate a broad range of frontier multimodal foundation models that support *video* or *multiple images* as input. Specifically, we evaluate eight series of **open-source models**, including InternVL2 [23, 25], Qwen2-VL [146, 164], LLaVA-NeXT [95], Pixtral [108], NVLM-D [33], LLaVA-NeXT-Video [84], Llama-3.2-Vision [40], Phi-3.5-Vision [1], InternVideo2 [149], and VideoLLaMA2 [26]. We also evaluate three series of **proprietary models**, including GPT-4o [117], Gemini-1.5 [53], and Claude-3.5 [4]. For open-source models, we prioritize the vLLM pipeline [81] for model inference; otherwise, we use the Transformers pipeline [156]. We use the official API service for proprietary models. For models without native video support, following VideoMME [51], we provide visual input using the maximum number of images that fits within the model's context window. §C.1 details the parameter settings and model configurations. We evaluate the models with both **Direct Answer** and **Chain-of-Thought** prompts (presented in Appendix C.3), which is adapted from the versions used in MMMU-Pro [166].

	Release	Breakdown Results (Test Set, CoT)				Overall (Validation)		Overall (Test)	
		Science	Healthcare	Human. & Social Sci.	Engineering	Direct	CoT	Direct	CoT
Human Performance (§3.4)									
Close-book		54.7	42.7	44.7	56.7			49.7	
Open-book		86.7	84.7	92.7	83.3			86.8	
Oracle		95.3	93.3	96.0	96.7			95.3	
Text-Only									
Qwen2-VL-72B	2024-09	–	–	–	–	26.3 ±1.8	28.1 ±1.2	–	–
Llama-3.2-90B-Vision	2024-09	–	–	–	–	27.7 ±1.2	26.4 ±0.9	–	–
InternVL2-76B	2024-07	–	–	–	–	24.4 ±0.7	25.9 ±1.0	–	–
GPT-4o	2024-08	53.4	51.6	51.3	50.7	49.7	52.1	48.5	51.9
Claude-3.5-Sonnet	2024-07	51.6	48.6	49.7	46.0	47.9	50.9	46.3	48.9
Gemini-1.5-Pro	2024-06	50.9	47.2	50.1	45.3	41.4	46.5	45.6	48.2
GPT-4o-mini	2024-07	49.9	44.7	47.3	44.0	43.2	46.8	44.5	46.5
Claude-3.5-Haiku	2024-10	49.8	47.4	46.4	48.5	43.0	47.3	43.2	48.5
Gemini-1.5-Flash	2024-09	50.4	44.3	44.7	41.5	42.2	44.3	41.8	45.3
Qwen2-VL-72B	2024-09	50.4 ±1.5	48.8 ±1.6	45.0 ±1.6	45.7 ±1.7	43.0 ±0.9	46.4 ±1.7	42.0 ±0.7	47.7 ±1.1
Llama-3.2-90B-Vision	2024-09	48.6 ±0.8	45.6 ±0.8	48.2 ±0.8	43.6 ±0.8	45.0 ±0.5	48.4 ±0.8	41.2 ±0.4	46.3 ±0.6
NVLM-D-72B	2024-10	45.5 ±0.4	42.7 ±0.3	46.7 ±0.3	40.4 ±0.3	39.7 ±0.2	42.3 ±0.4	40.5 ±0.2	43.4 ±0.2
InternVL2-76B	2024-07	47.2 ±1.0	44.2 ±1.1	41.6 ±1.1	38.9 ±1.1	41.4 ±0.6	42.8 ±0.9	39.7 ±0.6	42.9 ±0.7
Pixtral-12B	2024-09	46.6 ±0.3	43.0 ±0.3	40.8 ±0.3	37.5 ±0.3	44.3 ±0.2	43.6 ±0.3	42.9 ±0.1	41.9 ±0.2
Qwen2-VL-7B	2024-08	50.8 ±0.8	35.8 ±0.7	37.7 ±0.7	35.9 ±0.7	40.8 ±0.7	41.7 ±0.7	36.2 ±0.5	40.7 ±0.6
Llama-3.2-Vision-11B	2024-09	44.9 ±1.0	41.7 ±1.0	39.1 ±1.1	36.6 ±1.1	38.2 ±0.8	41.9 ±1.0	37.7 ±0.4	40.5 ±0.7
InternVL2-8B	2024-07	41.2 ±1.8	38.7 ±1.8	39.1 ±1.8	37.9 ±1.7	36.8 ±1.6	39.2 ±1.5	36.5 ±0.9	39.3 ±1.4
InternVideo2-8B	2024-08	41.4 ±1.1	39.1 ±1.1	37.7 ±1.1	35.9 ±1.1	35.2 ±0.9	37.0 ±1.2	37.4 ±0.6	38.5 ±1.4
Qwen2-VL-2B	2024-09	37.6 ±3.1	37.7 ±2.8	39.5 ±2.7	35.6 ±2.6	36.3 ±2.8	35.5 ±2.4	38.3 ±1.4	37.3 ±2.2
VideoLLaMA2-7B	2024-06	36.7 ±3.3	37.6 ±3.6	37.3 ±3.6	32.7 ±3.7	34.0 ±3.5	35.3 ±3.4	32.3 ±1.7	35.6 ±2.4
Phi-3.5-Vision	2024-08	36.0 ±1.9	32.7 ±1.7	33.8 ±1.9	33.4 ±1.9	28.8 ±2.0	33.5 ±2.1	31.3 ±0.9	34.2 ±1.5
LLaVA-NeXT-8B	2024-07	29.4 ±0.7	23.8 ±0.8	32.2 ±0.8	27.9 ±0.9	29.3 ±0.7	27.4 ±1.0	30.2 ±0.4	28.3 ±0.9
LLaVA-NeXT-Video-7B	2024-06	23.4 ±1.4	21.5 ±1.3	31.3 ±1.2	30.4 ±1.2	28.8 ±1.4	27.4 ±1.2	27.2 ±0.7	26.8 ±1.5

Table 2. Accuracy of different evaluated systems on the MMVU validation and test set using the prompting methods of Chain-of-Thought (CoT) and Direct Answer (Direct), respectively. We use the overall results of CoT on test set as the ranking indicator of model performance. For open-source models, we average the results over three runs. Underlined numbers in "Direct" column highlight that models perform worse with CoT reasoning compared to direct answering.

4.2. Automated Evaluation System Development

We use accuracy as the primary metric to evaluate model performance on MMVU. Foundation models often generate detailed and verbose responses, making automated evaluation challenging [112, 159]. Recent benchmark works either (1) apply rule-based methods to extract the final answer [51, 165, 166], which may lack precision and robustness, or (2) rely on proprietary LLM APIs to compare generated answers with ground truth [58, 100, 130, 151], which can introduce variability over time. To ensure a *reliable* and *consistent* evaluation, we employ Llama-3.1-8B fine-tuned on human-scoring data as the foundation for our evaluation system, due to its strong performance among 7B-level LLMs. We first construct a dataset of human-labeled scores. Specifically, we randomly sample 1,400 model responses from different evaluated models. Each response is annotated by a human evaluator following a CoT reasoning format: first identify the final answer in the model's response, then compare it to the ground-truth answer, and finally assign an accuracy score of 0 or 1. Appendix C.2 presents the dataset examples, along with the CoT scoring format. We then fine-tune the model using LoRA training for 5 epoches with the LLaMA-Factory pipeline [173] on this human-annotated dataset.

To validate its reliability, we conduct a human study on 300 random-sampled model responses (50 each from GPT-4o-mini, Gemini-1.5-flash, Pixtral-12B, InternVL2-7B, and Llama-3.2-11B-Vision, LLaVA-NeXT-8B). In comparison, the GPT-4o-mini-based evaluation system adapted from CharXiv [151] produce seven incorrect scoring; while the rule-based evaluation system adapted from MMMU [165] produce 19 incorrect scoring. Our evaluation system produces only four cases of incorrect scoring, demonstrating its strong reliability for MMVU evaluation.

Primary Error Type & Explanation
Visual Perception Error (16%): The model fails to accurately interpret spatial, temporal, or semantic aspects of visual information within a video. Additionally, it might "hallucinate", detecting objects or events that are not actually present in the video.
Misuse or Lack Domain Knowledge in Visual Perception (20%): The model fails to apply the domain-specific expertise required to accurately interpret specialized concepts or elements within the video. For example, in a medical video, it may identify objects but fail to recognize their technical terms or misunderstand their importance within the procedure being demonstrated. This limitation underscores a gap in the model's ability to integrate domain knowledge with visual perception effectively.
Logical Reasoning Error (9%): The model exhibits inconsistencies between its reasoning process and final answer, leading to self-contradiction.
Misuse or Lack Domain Knowledge in Reasoning (27%): The model fails to effectively recall and apply domain knowledge during its reasoning processes. For instance, when addressing questions over chemistry videos, it may fail to correctly apply relevant chemical equations, leading to errors in computing the reaction mass. This limitation underscores the model's inability to integrate domain knowledge into its reasoning processes effectively.
Heavy Reliance on Textual Information (16%): The model predominantly depends on textual information for problem-solving, especially when addressing multiple-choice questions, as it evaluates each option individually without leveraging the actual video content. Similar limitations have been observed in other multimodal benchmarks [51, 165]. This gap suggests future work in enhancing multimodal reasoning by more effectively incorporating non-textual content into the reasoning process.
Other Error (12%): This includes other errors, such as refusing to answer a question due to insufficient context or safety concerns, generating a response that exceeds the 512-character output limit, generating repetitive information, or making incorrect math computation.

Table 3. Explanations of common error types identified in MMVU, along with their frequency observed in Qwen2-VL-72B by humans. Due to space constraints, detailed case study examples for each error type can be found in Appendix D.

4.3. Main Findings

Table 2 presents the model performance. We observe a marked performance gap in models when restricted to a text-only setting; for instance, Qwen2-VL-72B experiences a performance drop from 46.4% to 28.1%, emphasizing the **critical role of visual perception in MMVU.** This disparity highlights the effectiveness of our quality control measures (§3.3). **MMVU presents substantial challenges for current multimodal foundation models.** Even the top-performing model falls well short of human expert performance. Specifically, GPT-4o achieves only 51.9% accuracy with CoT prompting, significantly lower than the 86.8% accuracy achieved by human experts in an open-book setting. Notably, while the state-of-the-art foundation model (*i.e.*, GPT-4o) has narrowed the performance gap with human experts in text-based expert-level reasoning on MMLU (88.7% vs 89.8% [60]) and image-based expert-level reasoning on MMMU (69.1% vs 82.6% [165]), the gap remains significantly larger on MMVU. This disparity underscores MMVU's critical role in advancing and evaluating foundation models' capabilities in video-based expert reasoning across specialized domains. As for open-source multimodal foundation models, they still lag behind the proprietary models. However, **the Qwen2-VL-72B and Llama-3.2-90B-Vision models come close to matching the performance of the leading proprietary models**, indicating that open-source models are making significant strides. Moreover, **the introduction of CoT reasoning generally lead to improved performance compared to direct outputting the answer.** However, we observe a significant performance drop in certain less-capable models, such as LLaVA-NeXT-series models. Our case studies reveal that these models often struggle to follow instructions accurately when generating CoT explanations, highlighting the critical importance of robust instruction-following capabilities for effective CoT reasoning.

5. Error Analysis and Case Study

To gain a deeper understanding of the capabilities and limitations of frontier models on MMVU, we perform comprehensive case studies and error analysis by both human and automated evaluation. The inclusion of expert-annotated reasoning rationales and domain knowledge for each example in MMVU facilitate a more effective analysis compared to datasets that provide only answers.

5.1. Human Evaluation

We focus on three top-performing open-source models, Qwen2-VL-7B and Llama-3.2-11B-Vision, and Pixtral-12B, for human evaluation. From the MMVU validation set, we randomly sample 50 correct cases and 100 error cases for each model. These cases are analyzed by the authors using ground-truth features (*i.e.* expert-annotated reasoning rationales and required domain knowledge) as references. Table 3 illustrates six primary error categories identified by human evaluators. To support future researchers in understanding the strengths and limitations of cutting-edge models on MMVU, we provide examples of both correct and incorrect model responses in Appendix D, accompanied by detailed human analysis for each example.

5.2. Large-scale Automated Error Analysis

Human-based error analysis provides deep insights but is time-consuming and lacks scalability, limiting the feasi-

Retrieval Setting	R@5	Llama-3.2-V	Qwen2-VL
–	–	41.9	41.7
Retriever: BM25			
Question	31.4	43.7 (+1.8)	45.9 (+4.2)
+ video desc.	32.7	43.3 (+1.4)	46.1 (+4.4)
+ response	41.2	45.3 (+3.4)	47.8 (+6.1)
+ video desc. & resp.	40.3	45.0 (+3.1)	47.2 (+5.5)
Retriever: GritLM-7B			
Question	42.3	46.1 (+4.2)	46.7 (+5.0)
+ video desc.	43.7	46.9 (+5.0)	47.7 (+6.0)
+ response	52.2	48.2 (+6.3)	49.1 (+7.4)
+ video desc. & resp.	51.9	48.4 (+6.5)	48.8 (+7.1)
Oracle	100.0	51.5 (+9.6)	53.2 (+11.5)

Table 4. CoT Performance of Llama-3.2-11B-Vision and Qwen2-VL-7B on the validation set across various RAG settings.

bility of frequent evaluations. To overcome these limitations and facilitate future research, we implement an automated error analysis protocol inspired by recent studies [109, 134, 160]. In brief, the multimodal evaluator (*i.e.*, GPT-4o) is provided with the expert-annotated reasoning rationales and required domain knowledge to improve the precision of error analysis. It is instructed to identify the main error step in the model's response that leads to the incorrect answer and categorize it into one of the six predefined error types based on prior human analysis. The implementation details of the protocol are provided in Appendix E. To assess the reliability of this protocol, we conduct a manual analysis of 100 examples, revealing that 92 of them are correctly analyzed by the protocol. We believe that future work can further refine this protocol, improving precision and leveraging open-source models to enhance transparency. Appendix D illustrates the distribution of error types among the top-6 performing open-source models on the validation set. A key limitation observed across models is their limitations in accurately recalling and applying domain knowledge during both visual perception and reasoning processes. This motivates us to explore RAG-based knowledge augmentation [6, 83] in the next section.

6. Knowledge Augmentation Analysis

Section 3.4 demonstrates that well-educated humans show significant performance improvements in open-book settings, where they have access to external information. Our goal is to investigate whether models can achieve similar gains through external knowledge augmentation.

6.1. RAG-Based Experiment Setup

We select two retrievers for RAG experiments: (1) BM25 [126], a widely-used sparse retriever and (2) GritLM-7B [111], the top-performing open-source retriever on the MTEB Retrieval Arena leaderboard [110]. We use the Wikipedia page corpus curated by MTEB [110] as the retrieval knowledge base. Following the setup in BRIGHT [133] for knowledge augmentation in theorem-based question answering, we explore a range of retrieval configurations using various query formulations: (1) the textual question, (2) the detailed video description generated by the evaluated foundation models (Appendix F.2 presents the used prompt), and (3) the model's original response collected from the main experiment. For each query type, we retrieve the top-5 Wikipedia pages and integrate their content into the CoT reasoning prompt (Appendix F.1), preserving up to first 200 words per page with paragraph boundaries maintained [107]. To investigate the headroom in RAG-based knowledge augmentation, we also test the oracle setting. In this setup, we include the ground-truth Wikipedia page content and supplement it with the top pages retrieved using the question as query. This approach ensures that the total number of input Wikipedia pages remains at least five for fair comparison.

6.2. Main Findings

As illustrated in Table 4, RAG-based knowledge augmentation generally enhances model performance, with higher retrieval recall correlating with improved outcomes. Notably, smaller models under RAG settings can surpass the performance of larger models within the same family. For example, with knowledge augmentation, Qwen2-VL-7B achieves an accuracy of 49.1%, outperforming Qwen2-VL-72B (46.4%) wihout knowledge augmentation. Additionally, incorporating model responses into query formation further improves retrieval results, whereas adding video descriptions does not bring much improvement. Our human analysis (Appendix F.3) suggests that the video descriptions generated by models often lack professional terminology, which likely contributes to less effective retrieval. We believe that future work could explore expanding the RAG setting to include multimodal retrieval [158, 170].

7. Conclusion

This work presents MMVU, a high-quality, multi-discipline benchmark designed to evaluate the expert-level reasoning capabilities of foundation models on specialized-domain videos. Our evaluation of 20 frontier multimodal foundation models demonstrates substantial progress, particularly among the latest open-source models. Nevertheless, a significant performance gap persists between the top-performing models and human experts. Our comprehensive error analysis, case studies, and investigation into RAG-based knowledge augmentation highlight persistent challenges, providing insights into enhancing foundation models' capacity for expert-level video understanding.

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